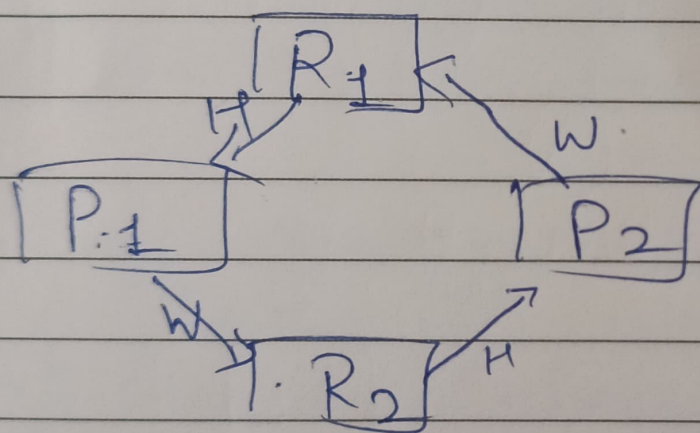


Unit 4:- Deadlock

→ It is a situation where two or more processes are unable to proceed because each process is waiting for the other Resources to release that resources.



Key Components to check whether it is a deadlock condition is there or not:-

- Mutual Exclusion
- No preemption
- hold & wait
- Circular way.

* Deadlock Prevention :-

→ It ensures that at least one of the four necessary conditions for deadlock never holds.

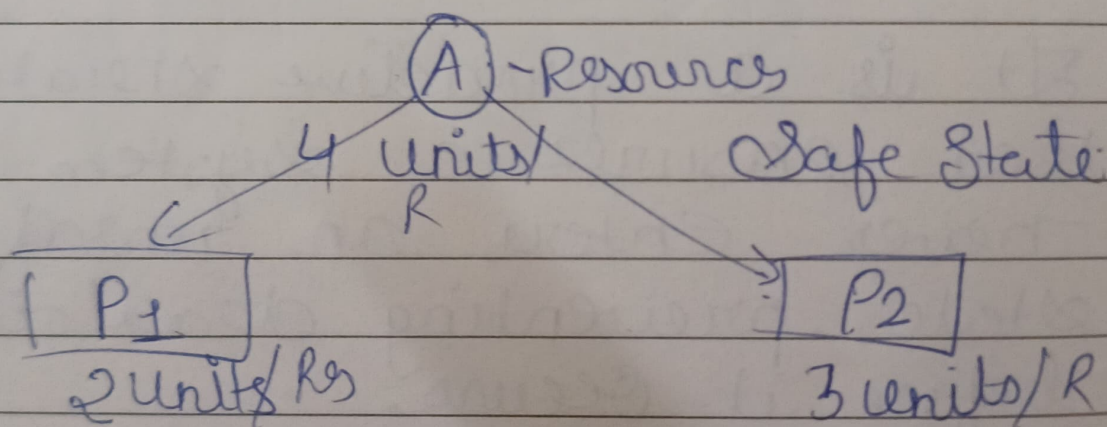
* Deadlock Avoidance :-

⇒ It is a proactive strategy that ensures a system never enters an unsafe/dead state, preventing deadlock before it occurs.

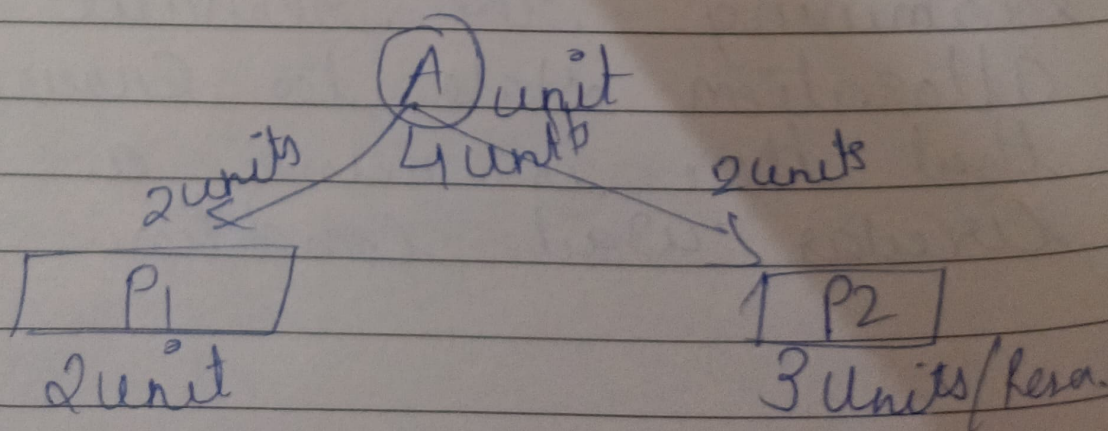
⇒ It works by dynamically examining the resource allocation state to ensure that there is never a circular wait condition.

* Key Concepts in Deadlock Avoidance -

1] Safe State :- A state where the OS can allocate resources to each process (up to its maximum) in a particular order without entering a deadlock.



2] Unsafe State :-



⇒ A state that might lead to a deadlock, the system can't guarantee that all processes can complete.

3]. Maximum Demand :- Processes must declare the maximum number of resources of each type they may ever need in advance.

* Banker's ~~Algo~~ Algorithm :-
(Deadlock Avoidance)

⇒ The Banker's Algorithm is a resource allocation & deadlock avoidance algorithm used by OS to safely manage limited resources.